CPU Scheduling Criteria, FCFS & SJF

Today's Learning Objectives

By the end of this lecture, students should be able to:

- Explain CPU scheduling.
- Understand scheduling criteria.
- Define Arrival Time, Burst Time, Completion Time, Turnaround Time, Waiting Time and Response Time.
- Draw a Gantt chart.
- Solve **FCFS** scheduling numericals.
- Solve **SJF (Non-Preemptive)** scheduling numericals.
- Calculate average waiting and turnaround time.

CPU Scheduling Algorithms — FCFS & SJF

Topics:

- Scheduling Criteria
- Scheduling Terminology
- Gantt Chart
- First-Come, First-Served (FCFS)
- Shortest Job First (SJF)
- Numerical Problems

Why Do We Need Scheduling Algorithms?

Suppose five processes are waiting:

Ready Queue

P1 P2 P3 P4 P5

↓ ↓ ↓ ↓ ↓

CPU Scheduler

↓

CPU

Question:
Which process should execute first?
There can be different answers depending on the scheduling algorithm.
Examples:

- FCFS → first process arrives first
- SJF → shortest CPU burst first
- Priority → highest priority first
- Round Robin → fixed time quantum

Scheduling Criteria

We evaluate scheduling algorithms using several criteria.

Main Criteria

1. **CPU Utilization**
2. **Throughput**
3. **Turnaround Time**
4. **Waiting Time**
5. **Response Time**

Important for Numericals ⭐

CT → TAT → WT → RT

CPU Utilization

Definition

CPU utilization represents the percentage of time the CPU is busy.

$$\text{CPU Utilization} = \frac{\text{CPU Busy Time}}{\text{Total Time}} \times 100$$

Example

CPU is busy for **90 ms** during a total period of **100 ms**.

$$\text{CPU Utilization} = \frac{90}{100} \times 100$$

=90%

Goal

Generally, we want **high CPU utilization**.

Throughput

Definition
Throughput = Number of processes completed per unit time

$$\text{Throughput} = \frac{Number\ of\ completed\ processes}{Total\ time}$$

Example

5 processes complete in 10 seconds.

$$Throughput = \frac{5}{10} = 0.5\ processes/sec$$

Simple Meaning
"How much work is completed in a given amount of time?"

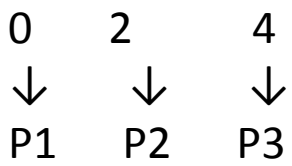
Important Scheduling Terms

Term	Meaning
AT	Arrival Time
BT	Burst Time
CT	Completion Time
TAT	Turnaround Time
WT	Waiting Time
RT	Response Time

Arrival Time (AT)

Arrival Time
The time at which a process enters the ready queue.
Example:

Process	Arrival Time
P1	0
P2	2
P3	4



“If P1 has AT = 0, it is available right from the beginning. If P2 has AT = 2, we cannot execute P2 before time 2.”

Burst Time (BT)

CPU Burst Time
The amount of CPU time required by a process.

Process	Burst Time
P1	5 ms
P2	3 ms
P3	2 ms

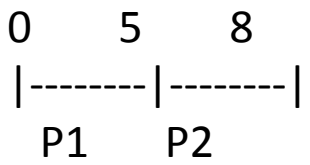
Meaning:

- P1 needs 5 ms CPU time.
- P2 needs 3 ms.
- P3 needs 2 ms.

Important
Burst Time ≠ Waiting Time

Completion Time (CT)

Completion Time
The time at which a process finishes execution.



Therefore:

- P1 CT = 5
- P2 CT = 8

Remember
CT = Time when process finishes

Turnaround Time

Formula ⭐
TAT = CT - AT
Meaning
Total time spent by a process in the system.
It includes:
Waiting Time + CPU Execution Time

Example
If:

- AT = 2
- CT = 10

Then:
TAT = 10-2=8

Waiting Time

Formula ⭐

$$\boxed{WT = TAT - BT}$$

Since:

$$TAT = CT - AT$$

we can also write:

$$\boxed{WT = CT - AT - BT}$$

Meaning

Time spent waiting in the ready queue.

Example

If:

- TAT = 8
- BT = 5

Then:

WT=8-5=3

Response Time

Formula

$$\boxed{RT = First\ CPU\ Start\ Time - AT}$$

Important Difference

Waiting Time can include multiple waiting periods.

Response Time measures how long the process waits before getting CPU for the **first time**.

Example

P1:

- AT = 2
- First CPU start = 5

RT=5-2=3

Numerical Solving Procedure

Step 1
Write the process table.
Step 2
Identify the scheduling algorithm.
Step 3
Determine execution order.
Step 4
Draw the Gantt chart.
Step 5
Calculate CT.
Step 6
Calculate TAT.
TAT=CT-AT
Step 7
Calculate WT.
WT=TAT-BT
Step 8
Calculate RT.
RT=First\ Start-AT
Step 9
Calculate averages.

FCFS Scheduling

First-Come, First-Served
Definition
The process that arrives first gets the CPU first.
It follows:
FIFO — First In, First Out
Ready Queue
P1 → P2 → P3 → P4
Execution:
P1 → P2 → P3 → P4

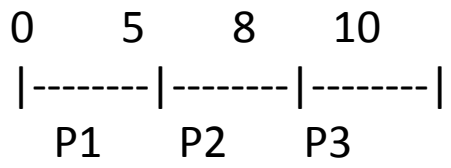
Type
Non-preemptive
Once a process gets the CPU, it keeps the CPU until it finishes or blocks.

FCFS Example 1

Given:
All processes arrive at time 0.

Process	AT	BT
P1	0	5
P2	0	3
P3	0	2

FCFS Gantt Chart



Completion Times

Process	CT
P1	5
P2	8
P3	10

FCFS Calculate TAT

Formula:
TAT=CT-AT
P1
TAT=5-0=5
P2
TAT=8-0=8
P3
TAT=10-0=10

Process	AT	BT	CT	TAT
P1	0	5	5	5
P2	0	3	8	8
P3	0	2	10	10

FCFS Calculate WT

Formula:
WT=TAT-BT
P1
WT=5-5=0
P2
WT=8-3=5
P3
WT=10-2=8
Final table:

Process	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	0	3	8	8	5
P3	0	2	10	10	8

Average TAT

$$ATAT = \frac{5 + 8 + 10}{3}$$

ATAT=7.66

Average Waiting Time

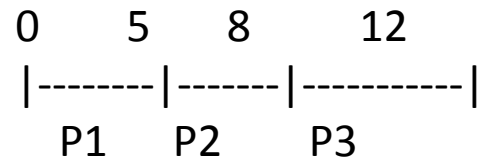
$$AWT = \frac{0 + 5 + 8}{3}$$

AWT=4.33

FCFS Example 2: Arrival Times

Process	AT	BT
P1	0	5
P2	1	3
P3	2	4

FCFS Example 2 Gantt Chart



Process	CT
P1	5
P2	8
P3	12

FCFS Example 2 Complete Solution

P	AT	BT	CT	TAT = CT−AT	WT = TAT−BT
P1	0	5	5	5	0
P2	1	3	8	7	4
P3	2	4	12	10	6

Average Turnaround Time:

$$ATAT = \frac{5 + 7 + 10}{3} \qquad \boxed{ATAT = 7.33}$$

Average Waiting Time:

$$AWT = \frac{0 + 4 + 6}{3} \qquad \boxed{AWT = 3.33}$$

FCFS Convoy Effect

FCFS can suffer from the:
Convoy Effect
Suppose:
P1 = 20 ms
P2 = 2 ms
P3 = 2 ms
P4 = 2 ms
FCFS:
P1 → P2 → P3 → P4

The short processes have to wait behind the long process.
Analogy
A long truck is driving in front of several cars on a single-lane road.



Everyone behind has to wait.
“This is one limitation of FCFS. A long CPU-bound process can make many short processes wait.”

SJF Scheduling

Shortest Job First
Definition
The process with the **shortest CPU burst time** is selected first.

Example:

Process	BT
P1	8
P2	3
P3	5
P4	2

Execution order:
P4 → P2 → P3 → P1
Because:
2 < 3 < 5 < 8

SJF Characteristics

SJF is usually:

- **Non-preemptive** when we specifically say **Non-Preemptive SJF**
- Based on **Burst Time**
- Designed to reduce average waiting time under the standard idealized assumptions

Important

Do not confuse:

SJF → non-preemptive

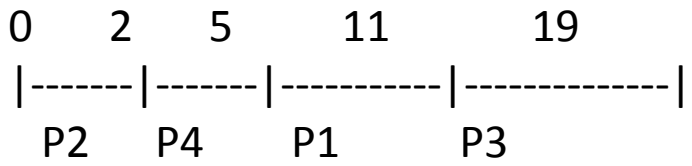
SRTF → preemptive version

SJF Numerical Example

Given:
All processes arrive at time 0.

Process	AT	BT
P1	0	6
P2	0	2
P3	0	8
P4	0	3

Step 1: Arrange by Burst Time
P2 = 2
P4 = 3
P1 = 6
P3 = 8
P2 → P4 → P1 → P3



P	AT	BT	CT	TAT	WT
P1	0	6	11	11	5
P2	0	2	2	2	0
P3	0	8	19	19	11
P4	0	3	5	5	2

Average Waiting Time

$$AWT = \frac{5+0+11+2}{4} \quad \boxed{AWT = 4.5}$$

Average Turnaround Time

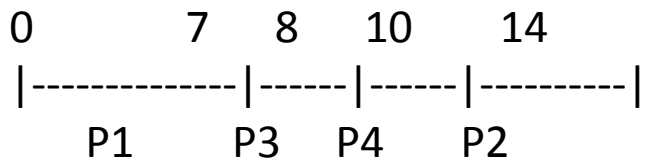
$$ATAT = \frac{11+2+19+5}{4} \quad \boxed{ATAT = 9.25}$$

SJF with Arrival Times

A process cannot be selected before it arrives.
Consider:

Process	AT	BT
P1	0	7
P2	2	4
P3	4	1
P4	5	2

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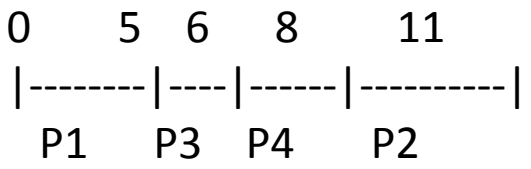


P1 → P3 → P4 → P2

SJF

Process	AT	BT
P1	0	5
P2	1	3
P3	2	1
P4	3	2

1. Gantt chart
2. CT
3. TAT
4. WT
5. Average WT
6. Average TAT



P	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	1	3	11	10	7
P3	2	1	6	4	3
P4	3	2	8	5	3

Average Waiting Time

$$AWT = \frac{0 + 7 + 3 + 3}{4} \quad \boxed{AWT = 3.25}$$

Average Turnaround Time

$$ATAT = \frac{5 + 10 + 4 + 5}{4} \quad \boxed{ATAT = 6}$$

Problem 1 — FCFS

Process	AT	BT
P1	0	6
P2	1	3
P3	2	4
P4	4	2

Problem 2 — FCFS

Process	AT	BT
P1	0	7
P2	0	10
P3	0	23
P4	0	4

Problem 3 — FCFS

Process	AT	BT
P1	0	4
P2	1	5
P3	2	3
P4	3	5
P5	4	6

Problem 4 — FCFS

Process	AT	BT
P1	4	4
P2	6	3
P3	0	2
P4	9	1
P5	5	3

Problem 1 — SJF

Process	AT	BT
P1	0	7
P2	1	2
P3	2	5
P4	3	1

Problem 2 — SJF

Process	AT	BT
P1	0	8
P2	2	5
P3	1	9
P4	3	2